Unlocking the potential of open language data as carriers of social and cultural information: The role of research infrastructures, data journals and training programmes to maximize reuse

Fišer, Darja
darja.fiser@inz.si
CLARIN ERIC, Netherlands, The

McGillivray, Barbara
barbara.mcgillivray@kcl.ac.uk
King's College London, GB

Frontini, Francesca
francesca.frontini@ilc.cnr.it
CLARIN ERIC, Netherlands, The

Brief abstract

This panel showcases the need for stronger collaboration between research infrastructures enabling data FAIR-ness, training programmes ensuring competent reuse of language data and data journals establishing rigorous review processes. This is essential to ensure data quality, relevance, and impact, maximising its potential for reuse in research, education and societal contexts.

Keywords

open language data, FAIR principles, research infrastructure, publishing, training

Panel abstract

Given that language is the carrier of social and cultural information as well as human knowledge, both the availability and importance of open language data have grown rapidly in DH research and practice in the past decades. However, the effort required to make and keep language data open and reusable on the one hand, as well as the knowledge required to reuse such data responsibly and reliably, are still not sufficiently recognized, especially across disciplinary boundaries. Insufficient attention is also paid to the importance of interoperability and standardisation of open language data in various modalities (text, speech, multimodal) and its critical role in addressing the fragmentation of the largely underfunded and understaffed DH research landscape, especially across language boundaries.

In this panel, we bring together DH scholars, trainers and practitioners from Europe, South Africa and Asia with extensive experience in creation and reuse of open language data from the perspectives of research, pedagogy, preservation and publishing. They will highlight the potentials and obstacles of textual and spoken open language data, and its metadata reuse, within and across disciplines. Special emphasis will be placed on the issues involved in language documentation and preservation, accessibility and reusability of language data in the light of the FAIR principles (Findable, Accessible, Interoperable, and Reusable) advocated by FAIRsFAIR, Research Data Alliance, as well as the CARE principles (Collective benefit, Authority to control, Responsibility, Ethics) for Indigenous Data Governance. Issues of data quality, relevance, potential for cross-disciplinary research and reuse for educational and societal purposes will be discussed. The panel will critically evaluate language data reuse and its interdisciplinarity, while exploring the necessary next steps.

The panel aims to demonstrate the critical need for a robust research curation and publication ecosystem, including research infrastructures such as CLARIN ERIC, which offers data, tools and services to support research based on language resources, for providing repositories, persistent identifiers, and standardised metadata models, mechanisms for knowledge transfer to support data findability, accessibility and reuse, and data journals and training programmes to document, evaluate, and disseminate datasets, while encouraging transparency and collaboration.

The panel features five contributions, each presenting real-world challenges and solutions around the creation, reuse, and curation of open language data. Topics include character encoding and peer-reviewed data publishing for Classical Chinese; access to secondary Holocaust research data; infrastructure-supported language documentation in South Africa; interdisciplinary collaboration using multimodal language data; and the role of data journals in promoting reuse and standardisation in the humanities. The contributions aim at showcasing the need for a stronger collaboration between research infrastructures enabling data FAIR-ness, training programmes to ensure competent reuse of open language data and data journals to establish rigorous review and assessment processes. Such efforts are essential to ensure data quality, relevance, and impact, ef-

fectively maximising its potential for reuse in research, education, and broader societal contexts. The panel will also highlight which next steps are needed for coordinated policy development and community-driven initiatives to advance the open language data ecosystem.

Abstracts from panelists

The Challenges of Using Classical Chinese Language Data in Digital Humanities

Youngim Jung, Korea Institute of Science and Technology Information

Jiwon Lee, Jeonbuk National University

Classical Chinese language data is crucial for DH research, but its use presents distinct challenges due to its linguistic, historical, and cultural intricacies. A key difficulty lies in managing the numerous character variants that evolved across historical periods and regions. Classical Chinese characters have traditional forms, still used in Taiwan and Hong Kong, while Simplified Chinese dominates in modern mainland China. Korea has developed unique adaptations of classical Chinese characters, known as Hanja. Disparities in character interpretation among these systems demand advanced algorithms for accurate encoding and cross-referencing (Heo & Cho, 2023).

To address these challenges, a project is conducted to encode Korean variants of Chinese characters into Unicode (Kim, 2021). Researchers have constructed lexicons for specific variants based on individual research objectives (Lee, 2024). However, accessibility and reusability of the language data still need to be improved.

This study explores the challenges of using character variation in Classical Chinese texts in an intertextuality study of the Sui and Tang Dynasties story, and describes a method to manage the variants in different texts. It highlights the importance of publishing and peer reviewing language data for DH. We will show how publishing language data can contribute to enhancing data quality, standardization, and interoperability (Jung et al, 2020). A formal review process can help identify inconsistencies, encourage the adoption of best practices, and provide a platform for critical evaluation, ultimately fostering a more reliable and comprehensive resource for researchers (Ahn et al, 2023).

Holocaust Histories and Access to Secondary Research Data

Jiři Kocian, Charles University

During the last ten years, providing online access to ever-increasing amounts of primary resources for historical research became a common practice. Holocaust studies, utilizing various sources ranging from statistical data to oral historical video interviews with Holocaust survivors, have played the role of a vanguard in the process of digitalization and global public accessibility. Academic and private researchers, educators, and the interested public have become familiar with the most common models and environments, which present them with primary data.

However, it still needs to be more common for the broader research public to utilize secondary data for their purposes. Over the last (probably even more than) 40 years of employing digital practices, the discipline has produced vast amounts of metadata and analytical and linguistic datasets. These contain invaluable information about the contexts, interlinkages between primary sources or the metadata sets, and additional layers of information ready to be recontextualized and interpreted, enriching each research with any specifics.

The presentation will discuss examples obtained from the experiences of the Malach Center for Visual History and EHRI-CZ node cooperation with operationalizing secondary Holocaust research data. It will focus on both modes of access and modes of presentation, aiming to identify a set of best practices that allow for easy finding, a clearer understanding of the nature of the data, and the benefits of secondary data usage.

Towards a science-literate and science-aware society: The contribution of language documentation and preservation

Juan Steyn, South African Centre for Digital Language Resources

SADiLaR is a South African national research infrastructure (RI) that is also a member of CLARIN . SADiLaR contributes towards a future where all twelve South African languages can effectively function in the digital space. Since its establishment, multiple projects and activities (SADiLaR, 2024a) have been initiated and as part of this panel contribution, a reflection will be offered on three main areas:

- First, specialisation projects run by SADiLaR nodes which speak to more traditional open language datasets and tool creation and the general challenges associated with linking disparate resources in African languages.
- Second, information I will share information on our SWiP project (SADiLaR, 2024b) creates opportunities for the broader language community to actively take part in the preservation and growth of African languages online through Wikipedia authorship training and subsequent creation of new content in African Languages.
- Finally, reflections on how SADiLaR benefited from access to open language data and infrastructure will be shared.

The harsh reality is that if no open language resources are available for a language so that its broader community can effectively engage with it in the digital domain, its future may only be that of a language that once was (Kornai, 2013). As an RI, we play our part in removing barriers to access for language communities to benefit from open language data and tools that are available but also more importantly for communities to be able to actively contribute and take ownership of the digital future of their languages.

Confronting the Challenges of Integrating Data Science and Multimodal Language Data: Insights from the Helsinki Digital Humanities Hackathon

Mikko Tolonen, University of Helsinki

Integrating data science and humanities poses significant challenges, often leading to projects that falter or are abandoned due to the complexities involved (Blevins 2016). Many initiatives attempt this integration only once, retreating when difficulties arise. The Helsinki Digital Humanities Hackathon confronts these challenges directly, leveraging years of practice to make interdisciplinary collaboration not just possible but fruitful. Rather than simplifying or circumventing the difficulties, the hackathon immerses participants in them, fostering an environment where sustained effort leads to meaningful integration (Ros & Oberbichler, 2020).

In this presentation, I will explore how the hackathon serves as an accelerated learning platform for the use of multimodal language data that does not shy away from the hard work of interdisciplinary collaboration. Participants from diverse backgrounds engage in project-based learning, working in teams to formulate research questions, design methodologies, and interpret results collectively, often around language data of different types. This hands-on approach helps participants develop a shared language and understanding, which is crucial for genuine collaboration between data science and humanities (Rockwell and Sinclair 2012).

Our experience with the Helsinki Digital Humanities Hackathon shows that, through dedication and iterative practice, the integration of data science and humanities and social science is not only possible but can lead to innovative research outcomes. This approach provides valuable insights for others seeking to bridge these disciplines and underscores the importance of perseverance in overcoming interdisciplinary challenges (Matres, Oiva & Tolonen 2018).

A publishing perspective: the role of data journals in language data reuse

Barbara McGillivray, King's College London

The growing prominence of language datasets in digital humanities research and practice in recent years has been facilitated by the increasing availability of digital collections, advances in data-intensive techniques, the emphasis on accessibility and preservation of intangible cultural heritage, and the development of infrastructure supporting the sharing, storage, and long-term sustainability of research data (Ma 2024). This evolving landscape has sparked rising interest in platforms that facilitate the publication of data papers, driven by a focus on how data are created, shared, cited, and reused (Larrousse & Gray 2021).

Data papers are peer-reviewed articles that not only describe specific datasets, but also emphasise their potential for reuse (Garcia-Garcia et al. 2015). Data journals play a critical role in enhancing data reuse by providing dedicated venues for datasets to be systematically documented, evaluated, and disseminated, encouraging transparency and collaboration. The rise of data journals also demonstrates the need for robust infrastructures, such as repositories, persistent identifiers, and standardised metadata models, to support data accessibility and reuse.

This presentation will focus on the Journal of Open Humanities Data and its efforts to promote language data reuse in DH, which has been shown to increase the datasets' visibility and impact (McGillivray et al. 2022). It will also explore the challenges and opportunities faced by data journals in promoting the reuse of data in DH, including issues of accessibility, ethics, peer review, and models of standardised data sharing practices, with a particular focus on language data.

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